Q1. What is a destructor in C++?

Answer: Special member function which deallocates memory

Marks: 1/1

Q2. Difference between constructor and destructor?

Answer: Constructor initializes object; Destructor destroys object

Marks: 1/1

Q3. Why are destructors important?

Answer: They delete dynamically allocated memory and files

Marks: 1/1

Q4. Can a class have multiple destructors?

Answer: No, destructor cannot be overloaded

Marks: 1/1

Q5. Why is operator overloading used?

Answer: Improves code readability

Marks: 1/1

Q6. Which operators cannot be overloaded?

Answer: :: . .* ?:

Marks: 1/1

Q7. Purpose of this pointer

Answer: To refer the current object

Marks: 1/1

Q8. Shallow copy vs Deep copy

Answer: Shallow copies address; Deep copies values

Marks: 1/1

Q9. Why user-defined copy constructor?

Answer: When a class uses dynamic memory

Marks: 1/1

Q10. Order in inheritance

Answer: Base->Derived constructor, Derived->Base destructor

Marks: 1/1

Q11. Role in dynamic memory allocation

Answer: Constructor allocates memory; Destructor releases it

Marks: 1/1

Q12. Program using constructor, destructor and operator overloading

Answer: Number class program provided

Marks: 1/1

Q13. Difference between C and C++

Answer: C is a procedural language. C++ is a multi-paradigm language.

Marks: 0.8/1

Q14. Call by Value vs Call by Reference

Answer: Call by value passes values whereas reference uses memory address.

Marks: 0.8/1

Q15. Class vs Object

Answer: Class is a blueprint while object is an instance of a class.

Marks: 0.9/1

Q16. What is Inheritance?

Answer: Inheriting properties from parent to child class. Types: Single, Multiple, Multilevel, Hierarchical, Hybrid.

Marks: 0.9/1

Q17. What is Polymorphism?

Answer: One function multiple actions. Compile-time resolved during compilation; Run-time during execution.

Marks: 0.9/1

Q18. Struct vs Class

Answer: Didn't answer.

Marks: 0/1

Q19. Pointer and Null Pointer

Answer: Pointer refers to memory address. Null pointer points to nothing.

Marks: 0.9/1

Q20. Stack vs Heap Memory

Answer: Stack manages local variables automatically. Heap is used for dynamic allocation.

Marks: 0.9/1

Q21. Function Overloading

Answer: Function name with different parameters or data types.

Marks: 0.9/1

Q22. Array vs Linked List

Answer: Array stores elements sequentially. Linked list stores elements in individual blocks.

Marks: 0.9/1

Q23. What is a constructor in C++?

Answer: Special member function called automatically when object is created

Marks: 1/1

Q24. Characteristics of a constructor

Answer: Called automatically, no return type, can be overloaded

Marks: 1/1

Q25. What is a default constructor?

Answer: Constructor with no parameters

Marks: 1/1

Q26. What is a parameterized constructor?

Answer: Constructor with parameters

Marks: 1/1

Q27. What is a copy constructor?

Answer: Initializes an object using another object

Marks: 1/1

Q28. When is a copy constructor called?

Answer: When an object is initialized from another object

Marks: 1/1

Q29. What is constructor overloading?

Answer: Same constructor name with different parameters

Marks: 1/1

Q30. Can a constructor return a value?

Answer: No, it is used to initialize objects

Marks: 1/1

Final Score: 27.9 / 30

Percentage: 93.00%